

GNSS-aided INS for land vehicle positioning and navigation

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Stockholm 2007

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TRITA-EE 2007:066

ISSN 1653-5146

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Abstract

This thesis begins with a survey of current state-of-the art in-car navigation systems. The pros and cons of the four commonly used information sources – GNSS/RF-based positioning, vehicle motion sensors, vehicle models and map information – are described. Common filters to combine the information from the various sources are discussed.

Next, a GNSS-aided inertial navigation platform is presented, into which further sensors such as a camera and wheel-speed encoder can be incorporated. The construction of the hardware platform, together with an extended Kalman filter for a closed-loop integration between the GNSS receiver and the inertial navigation system (INS), is described. Results from a field test are presented.

Thereafter, an approach is studied for calibrating a low-cost inertial measurement unit (IMU), requiring no mechanical platform for the accelerometer calibration and only a simple rotating table for the gyro calibration. The performance of the calibration algorithm is compared with the Cramr-Rao bound for cases where a mechanical platform is used to rotate the IMU into different precisely controlled orientations.

Finally, the effects of time synchronization errors in a GNSS-aided INS are studied in terms of the increased error covariance of the state vector. Expressions for evaluating the error covariance of the navigation state vector are derived. Two different cases are studied in some detail. The first considers a navigation system in which the timing error is not taken into account by the integration filter. This leads to a system with an increased error covariance and a bias in the estimated forward acceleration. In the second case, a parameterization of the timing error is included as part of the estimation problem in the data integration. The estimated timing error is fed back to control an adjustable fractional delay filter, synchronizing the IMU and GNSS-receiver data.

Acknowledgements

First of all, I would like to express my deepest gratitude to my advisor, Professor Peter Händel, for his ideas, inspiration and enormous support. I look forward to working with you for another couple of years!

I would like to thank my colleagues at "plan 4" for making work a pleasure. To my friends, who have repeatedly asked me what a PhD student actually does and what I am working on and, though they may not have fully understood my answers, still support me. Put simple, the work of a PhD student can be summarized as follows: Choose a topic (in my case land vehicle navigation), read one hundred papers on it, write a new paper with a couple of amendments so that the next person in line will have to read one hundred and one papers, present your results at a conference in a carefully chosen location and, lastly, iterate the process several times. Thanks for bringing a lot of joy and fun into my life.

Finally, and most importantly, I would like to thank my mother, Margareta, and my father, Rolf, for letting me as a child bring home and take apart all the old televisions and stereos I could find - that's how it all started. I owe it all to you. To my brother, Elias, and my half-sister, Julia, I love you the most!

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Part I

Introduction

Introduction

In-car navigation involves three distinguished processes: estimation of the vehicles position and velocity relative to a known reference, path planing, and route guidance. The first capability, positioning, is essential for successful path planing and route guidance capability. Nowadays, the area of high-performance positioning systems and methods is well developed. The challenge is to develop high-performance system solutions using low-cost sensor technology. This is the topic of the thesis, consisting of the following five papers.

Paper A: I. Skog and P. Händel, "State-of-the art and future in-car navigation systems – a survey", submitted to *IEEE Transactions on Intelligent Transportation Systems*.

Paper B: I. Skog and P. Händel, "A low-cost GPS aided inertial navigation system for vehicle applications", in *Proc. EUSIPCO 2005, (Antalya, Turkey), Sept. 2005*.

Paper C: I. Skog, A. Schumacher and P. Händel, "A Versatile PC-Based Platform For Inertial Navigation", in *Proc. NORSIG 2006, (Reykjavik, Iceland), June. 2006*.

Paper D: I. Skog and P. Händel, "Calibration of a MEMS inertial measurement unit", in *Proc. XVII IMEKO World Congress, (Rio de Janeiro, Brazil), Sept. 2006*.

Paper E: I. Skog and P. Händel, "Time synchronization errors in GPS-aided inertial navigation systems", submitted to *IEEE Transactions on Intelligent Transportation Systems*.

1 Contributions of the Thesis

The contributions in this thesis appears in terms of five papers, devoted to different areas associated with the development of low-cost in-car navigation solutions. An introduction to land vehicle navigation is provided in paper A, written as a survey of the current state-of-the art in-car navigation technology; to mediate a

understanding of the limitations and problems associated with the current in-car navigation systems. The remaining four papers make contributions to the following topics.

- *Development of versatile navigation platforms.* Papers B and C, presents the construction of a GNSS aided INS platform, into which further sensors such as a camera, wheel-speed encoder etc., are easily incorporated.
- *Calibration of low-cost IMUs.* The main contribution in paper D is the proposed simplified method to calibrate low-cost IMUs, together with the derivation of the Cramér-Rao bound for the standard calibration method, where a turn-table is used to rotate the IMU into different orientations.
- *Time synchronization in GNSS aided INSs.* Paper E deals with the problem of time synchronization in a GNSS aided INS. Expressions for the increased error covariance of the system, due to the synchronization error is derived. A method to compensate for the time synchronization error is proposed.

The papers are summarized in the following sections.

Paper A: State-of-the art and future in-car navigation systems – a survey

A survey of the information sources and information fusion technologies used in the current in-car navigation systems is presented. The pros and cons of the four commonly used information sources — GNSS/RF-based positioning, vehicle motion sensors, vehicle models and map information — are described. Common filters to combine the information from the various sources are discussed. A prediction of possible tracks in the further development of in-car navigation systems concludes the survey.

Paper B: A low-cost GPS aided inertial navigation system for vehicle applications

In this paper an approach for integration between GPS and inertial navigation systems (INS) is described. The continuous-time navigation and error equations for an earth-centered earth-fixed INS system are presented. Using zero order hold sampling, the set of equations is discretized. An extended Kalman filter for closed loop integration between the GPS and INS is derived. The filter propagates and estimates the error states, which are fed back to the INS for correction of the internal navigation states. The integration algorithm is implemented on a host PC, which receives the GPS and inertial measurements via the serial port from a tailor made hardware platform, which is briefly discussed. Using a battery operated PC the system is fully mobile and suitable for real-time vehicle navigation. Simulation results of the system are presented.

Paper C: A Versatile PC-Based Platform For Inertial Navigation

A GPS aided inertial navigation platform is presented, into which further sensors such as a camera, wheel-speed encoder etc., can be incorporated. The construction of the platform is described and an introduction to the sensor fusion approach is given. Results from a field-test is presented, indicating which error sources that needs to be modelled more accurately.

Paper D: Calibration of a MEMS inertial measurement unit

An approach for calibrating a low-cost IMU is studied, requiring no mechanical platform for the accelerometer calibration and only a simple rotating table for the gyro calibration. The proposed calibration methods utilize the fact that ideally the norm of the measured output of the accelerometer and gyro cluster are equal to the magnitude of applied force and rotational velocity, respectively. This fact, together with model of the sensors is used to construct a cost function, which is minimized with respect to the unknown model parameters using Newton's method. The performance of the calibration algorithm is compared with the Cramér-Rao bound for the case when a mechanical platform is used to rotate the IMU into different precisely controlled orientations. Simulation results shows that the mean square error of the estimated sensor model parameters reaches the Cramér-Rao bound within 8 dB, and thus the proposed method may be acceptable for a wide range of low-cost applications.

Paper E: Time synchronization errors in GPS-aided inertial navigation systems

The effects of time synchronization errors in a GPS-aided inertial navigation system (INS) are studied in terms of the increased error covariance of the state vector. Expressions for evaluating the error covariance of the navigation state vector — given the vehicle trajectory and the model of the INS error dynamics — are derived. Two different cases are studied in some detail. The first case considers a navigation system in which the timing error is not included in the integration filter. This leads to a system with an increased error covariance and a bias in the estimated forward acceleration. In the second case, a parameterization of the timing error is included as a part of the estimation problem in the data integration. The estimated timing error is fed back to control an adjustable fractional delay filter, synchronizing the inertial measurement unit (IMU) and GPS-receiver data. Simulation results show that by including the timing error in the estimation problem, almost perfect time synchronization is obtained and the bias in the forward acceleration is removed. The potential of the proposed method is illustrated with tests on real-world data that are subjected to timing errors. Moreover, through an observability analysis, it is shown that the timing error is observable for all trajectories that include turns or non-zero accelerations.

2 Related papers not included in the thesis

The following two papers have not been included, even though partly related to the work described in the thesis.

Paper F: J. Rantakokko, P. Händel, F. Eklöf, B. Boberg, M. Junered, D. Akos, I. Skog, H. Bohlin, F. Neregård, F. Hoffmann, D. Andersson, M. Jansson, and P. Stenumgaard, "Positioning of emergency personnel in rescue operations – possibilities and vulnerabilities with existing techniques and identification of needs for future R&D", *Technical report, Royal Institute of Technology, Stockholm, Sweden*.

Paper G: P. Händel, Y. Yao, N. Unkuri, and I. Skog, "A framework for moose early warning driver assistance systems", *Technical report, Royal Institute of Technology, Stockholm, Sweden*.